

General Discussion

(Draft, 2026-08-13; revised 2026-08-18 after the five-way cross-check against the Introduction, Chapters 1–3 and Appendix G. Evidence for the perspectives in Appendix G. Figures and cross-references to be renumbered at thesis assembly.)

D.1 Three questions, answered in order

This thesis asked how the understory microclimate of temperate deciduous forests can be modelled from remote sensing, given that each sensor sees only part of the canopy. The General Introduction split that problem into three questions to be addressed in order, and each answer narrows the next question.

Chapter 1 answered the attribution question. Leaf quantity, read as leaf area and cover together, governs the realised $\Delta T_{\max}$ buffering, and both levers steepen as the canopy closes. In the most open stands every effect falls below 0.03 °C, too small to rank. The vertical arrangement of foliage is second order. Outside those open stands the real-versus-uniform contrast stays at or below the cover step, and its realised effect is about 0.05 °C. Chapter 1 carries no field validation of its own. Chapter 3 supplies it. Over the 53 loggers the simulated and observed buffering agree at $r = 0.79$ on the coupling slope and $r = 0.63$ on the daytime offset. The model therefore recovers the ordering of the plots on the buffering gradient better than the amplitude of their offsets. If quantity governs the buffering, a two-dimensional sensor might suffice. Chapter 1 quantified the accuracy it must reach. A uniform 23 % reduction of the retrieved leaf area is the shift the optimised extinction coefficient implies. It warms the simulated $\Delta T_{\max}$ by 0.24 °C, and by 0.48 °C in dense canopy (Chapter 3, Appendix B). The plot ranking survives it. The scale therefore matters for the absolute coupling and not for the ordering, so an optical proxy must keep its leaf-area error well below 23 % wherever the absolute offset is used. That is the regime in which optical retrievals saturate.

Chapter 2 answered the consistency question. Sentinel-2 LAI saturates against the LiDAR reference in dense canopy, with a ceiling near 6–7 m² m⁻² where LiDAR reaches 10–15, and overshoots it in sparse canopy. The divergence is one of dynamic range rather than of central tendency. Regression slopes remain well below unity (0.25–0.80 across configurations) after every correction. The site means are comparable once the LiDAR extinction coefficient is set to $k = 0.65$, the value adopted for inter-sensor work. Chapter 1 keeps $k = 0.5$. The LiDAR integration depth at which the two products agree best, which we termed the effective optical depth d_{opt} , is 6–10 m (7 m across sites), against canopy tops of 17–32 m. It is an empirical proxy for the canopy thickness that dominates the Sentinel-2 signal, not a measured penetration depth. The third component was horizontal. Agreement declined with within-pixel canopy-height heterogeneity at all three sites, and the residual errors after both corrections remained largest where canopy tops are discontinuous. A single fixed d_{opt} is therefore an averaging that holds for closed, homogeneous canopies, and it marks the edge of the domain over which consistency was established. The chapter claimed consistency rather than accuracy, and it ended by proposing the optimised product as an operational input for microclimate modelling. Chapter 3 inherited that proposal, together with the question the General Introduction had left open: whether better agreement in leaf area

propagates into better simulated microclimate.

Chapter 3 answered the forcing question by testing that proposal. The best product depended on the canopy archetype. In dense canopy Sentinel-2 saturates: only the products carrying the LiDAR magnitude rank the plots (coupling slope $r = 0.72$ for LiDAR, 0.68 for the combination, 0.32 for raw Sentinel-2). In mid-density canopy the two cannot be separated on 12 plots: Sentinel-2 ranks at $r = 0.60$ against 0.67 on the slope and 0.75 against 0.69 on the daytime offset, both differences including zero. In the open gaps the LiDAR leaf area is near zero, which the point cloud confirms, while Sentinel-2 retrieves 4 to 5 $\text{m}^2 \text{m}^{-2}$; LiDAR is the reliable product there. Pooled over all plots, Sentinel-2 appears better ($r = 0.86$ against 0.79); the advantage is a composition effect of the archetypes. The dense-canopy conclusion is model-free (within-dense $r = -0.60$) and survives the robustness checks. The LiDAR leaf area is invariant to point density, down to a tenth of the native sampling (Spearman $\rho \geq 0.99$). Sentinel-2 adds its seasonality where the canopy does not saturate.

D.2 The canopy-density gradient across the three chapters

The three chapters describe one canopy-density gradient. Chapter 1 finds the leaf-quantity lever steepest in its densest archetype. Chapter 2 places the optical saturation near a 6–7 $\text{m}^2 \text{m}^{-2}$ ceiling, reached where the LiDAR estimate still climbs to 10–15. Chapter 3 shows that the Sentinel-2 ranking deficit grows with the fraction of leaf area below the effective optical depth ($r = -0.71$ across the dense archetypes). The three chapters do not share one extinction coefficient ($k = 0.5$ in Chapter 1, 0.65 in Chapters 2 and 3), so the numerical alignment should not be over-read. The mechanism is shared: radiation extinction accumulates with leaf area, so an optical sensor cannot see through the layers that buffer the understory most. In the dense Blois plots most of the leaf area sits below the effective optical depth and is invisible from above, so the Sentinel-2 magnitude saturates in the stands where the buffering is largest. Independent voxel-based simulations support this mechanism: Sentinel-2 near-infrared reflectance responds to the vertical position of the foliage density peak, and this sensitivity weakens as the foliage spreads vertically (Fujiwara et al., 2026).

The Chapter 1 sensitivities show the same geometry. Its leaf-area step is steepest in the densest archetype, and the two directions are not mirror images there (0.205 against 0.172 $^{\circ}\text{C}$), so the effect steepens as the canopy closes. The cover step is steeper still, 0.265 $^{\circ}\text{C}$ per 10 points, and Chapter 1 does not rank the two against each other. That geometry belongs to the sensitivities, not to the model’s skill. The simulation ranks the dense and the sparse halves of the 53 loggers equally well, at $r = 0.78$ and 0.82 on ΔT_{max} . Chapter 3 revises the reading of the open plots. The open plots are genuine gaps. Their LiDAR point clouds are almost all ground returns, so the near-zero LiDAR leaf area is correct. Sentinel-2 retrieves 4 to 5 $\text{m}^2 \text{m}^{-2}$ of a leaf area that is not there. The Sentinel-2 signal in the open therefore reflects the understory and soil background, not a tree canopy, and Sentinel-2 is the least reliable product there. Where the canopy is this sparse the simulated microclimate is only weakly sensitive to the forcing, so the open plots contribute little to the choice of sensor.

D.3 Inter-sensor consistency versus microclimate skill

The most consequential cross-chapter result is that inter-sensor consistency and microclimate skill are two objectives, and optimising the first can degrade the second. Accuracy here means fidelity to the retrieval objective, inter-sensor consistency (§D.5); with that gloss, The General Introduction posed it as an open, empirical question, and the chapters answered it three times. First, Chapter 1 showed that the coupling is calibrated to a scale. A uniform 23 % reduction of the LiDAR leaf area warms the simulated offset by 0.24 °C without disturbing the plot ranking (Chapter 3, Appendix B). An optical proxy must therefore stay well below that error wherever the absolute offset is used. That is the direction and order of the $k = 0.65$ rescaling that maximises inter-sensor consistency in Chapter 2. Second, Chapter 2 produced the most inter-sensor-consistent retrieval of the thesis, a site-specific PROSAIL configuration. It reduces the RMSE against the depth-truncated LiDAR LAI by 1.2 m² m⁻² on average and recentres the bias at two sites (a +0.7 m² m⁻² residual remains at Blois). At Blois, the site of Chapter 3, the selected LAI prior is the truncated LiDAR distribution itself, so the product was tuned to the upper 6 m of the canopy. Third, Chapter 3 found that this same product drives the dense-canopy microclimate worse than the untuned retrieval it was meant to improve on (dense coupling slope $r = 0.09$ against 0.32). In the model-free test its correlation with the observed buffering is sign-inverted ($r = +0.14$, where LiDAR gives -0.60). The same geometry sets the dense deficit directly: the Sentinel-2 ranking deficit grows with the fraction of leaf area buried below the effective optical depth ($r = -0.71$ across the dense archetypes).

The optimisation objective determines the outcome. Inter-sensor consistency and functional fidelity are two objective functions, and a retrieval tuned to match a saturating signal inherits the saturation. Chapter 2 proposed the optimised product as “an operational input” for microclimate modelling. Chapter 3 tested the proposal: in dense canopy the forest-tuned product is a worse driver than the raw retrieval (coupling slope $r = 0.09$ against 0.32). The two chapters are consistent. Chapter 2 built the best product the optical route allows. Chapter 3 shows that the constraint is upstream of the tuning, in the retrieved per-pixel leaf area. A random forest on the raw reflectance recovers dense ranking that the retrieval discards, at $\Delta T_{\max}$ $R^2 = 0.31$ on the ten bands and 0.48 on the three the inversion reads. The retrieved leaf area gives 0.02 there and LiDAR 0.42.

D.4 The LiDAR contribution: magnitude rather than vertical detail

Two statements in this thesis look contradictory until a distinction is made. Chapter 1 concludes that the vertical detail of ALS contributes little to the realised buffering signal. Replacing each real profile by its archetype mean moves $\Delta T_{\max}$ by 0.01 to 0.07 °C, and the realised profile effect is about 0.05 °C. A model-free test on the 53 loggers agrees. Adding the vertical profile to leaf area, cover and height raises the explained variance in the observed offset by 0.013 (permutation $p = 0.034$), and by 0.003 on the coupling slope ($p = 0.26$). Maximum height stays within 0.001 °C per metre of zero in every archetype, its widest interval bound being 0.003 °C. That figure is a lower bound, since the aerodynamic pathway is held fixed. The direction the General Introduction hypothesised, that foliage concentrated high in the canopy buffers more, holds

over most of the density range but as a second-order lever, and reverses at the dense end. Chapter 3 concludes that LiDAR is indispensable in dense canopy. Both hold, because they concern different properties of the instrument. The microclimate model needs the LiDAR magnitude because it does not saturate: the optical retrieval loses the between-plot ranking throughout the dense stratum, well before its $6\text{--}7\text{ m}^2\text{ m}^{-2}$ ceiling. The three-dimensional detail of ALS serves here mostly to measure leaf quantity without saturating. Canopy height plays a third role: negligible as a physical lever, it is a serviceable classifier of the density regime that organises Chapter 3, cheap to obtain wall-to-wall (Lang et al., 2022; Potapov et al., 2021; Schwartz et al., 2023).

The distinction also closes the question the General Introduction inherited from Bouwen (2025). That study attributed part of its under-buffering to the parametric beta profile it had to assume and recommended substituting measured LiDAR profiles. We made the substitution, in a deciduous canopy, and isolated the profile at matched leaf area and height: its realised contribution is small, and the warm bias and compressed amplitude persist unchanged (§D.6). The assumed profile was not the cause of the under-buffering; the one-dimensional column is.

D.5 Working without ground truth

No field LAI was used in this thesis, at any site, in any chapter, and none concurrent with the acquisitions was available. The argumentative structure was built to survive this. Three mutually independent supports replace the missing reference. The first is mechanistic attribution (Chapter 1), its safeguards were physical parameters fitted to no microclimate measurement, and perturbations centred on each plot’s own measured canopy. The second is inter-sensor consistency (Chapter 2), which quantifies agreement between two imperfect instruments without electing either as truth. The reference instruments themselves disagree in closed forests (Woodgate et al., 2015), so a field campaign would have relocated the problem rather than solved it. The third is functional constraint (Chapter 3), in which 53 temperature loggers, independent of both the model and the LAI retrievals, judge LAI products by what they are for. Each support has a named limitation. The attribution’s ranking is corroborated model-free, but its magnitudes are model-internal, and only the ranking was claimed to transfer. The consistency analysis carries a bounded prior circularity ($\leq 17\%$ of the reported RMSE reduction), an unbounded scoring circularity, and a reference-driven d_{opt} selection. The constraint is single-site, and its spatially non-independent loggers make the reported R^2 optimistic. The thesis therefore claims rankings and regime assignments, not calibrated effect sizes. It reports ranking, dispersion, and bias separately throughout, with a model-free leaf-area test as a fourth axis in Chapter 3. The axes disagree too often to be summarised by one number. The same discipline applies to the vapour-pressure-deficit offsets that Chapter 1 computed: they are model-internal, unvalidated by the loggers, and were carried only as a robustness check on the temperature result.

The point clouds settle one discrepancy between chapters directly. On the open plots the LiDAR leaf area differs by up to a factor of three between the archetype clustering of Chapter 1 and the forcing of Chapter 3. The two handle the near-ground returns differently. The most open plots are almost all ground returns, so the near-zero Chapter 3 value is correct and the Chapter 1 clustering overestimates leaf area there. The

archetype assignment is unaffected, since the eight open plots are the eight sparsest by either measure, so only the open-canopy magnitude differs, not the ranking.

D.6 Simulation defects common to all forcings

A recurring pattern across Chapters 1 and 3 sets the agenda for what follows this thesis. However the canopy is described, the simulation shares four defects. A warm bias survives every forcing tested, from +0.30 °C under the fixed LiDAR leaf area to +0.72 °C under the forest-tuned Sentinel-2 retrieval. The between-plot amplitude is compressed, to pooled dispersion ratios of 0.38 on the slope and 0.23 on the offset in Chapter 3. The compression sits in the gaps: in dense canopy the simulated spread matches the observed (1.03 and 0.79). The nighttime warm bias is +0.14 to +0.60 °C in Chapter 3, which Chapter 1 did not examine. Skill degrades on the hottest days, where predictions matter most. These defects move together when the forcing changes, but they do not close. Every forcing change tested so far moved leaf area down, the 23 % cut and an optical truncation, both in Chapter 3, and both worsened them. No chapter tested a higher leaf-area scale ($k < 0.5$, or a clumping correction), the direction that could reduce the warm bias inside the present column. Subject to that untested direction, the margin of progress has shifted from the forcing to the model. The one-dimensional column omits the sub-pixel gaps that structure the residual, together with edge effects, lateral advection, and cold-air drainage (Reek et al., 2025). The omission is largest on the amplifying tail of the gradient, the open plots that run warmer than the open air. Part of the measured gap there is protocol rather than physics: non-aspirated shields read warm in open, sunlit, low-wind stands (Terando et al., 2017), and the loggers sit at 1 m against a 1.5 m reference.

The nighttime side deserves one further observation, because an exploratory analysis of the logger archive (Appendix G, §G.8) changes how we read it. Nighttime $\Delta T_{\min}$ couples to LiDAR leaf area almost as strongly as daytime $\Delta T_{\max}$, with the opposite sign ($r = +0.83$ against -0.89). Closed canopy holds the night at the reference temperature while open stands fall 2 to 4 °C below it. That mirrors the daytime pattern, a bilateral amplitude compression under one structural control. Chapter 3 treated the night as a sensor-agnostic regime: the daytime reversal dissolves, both products rank the plots comparably (pooled $\Delta T_{\min}$ R^2 0.62–0.69), and the model runs +0.1–0.6 °C warm. The correlation above suggests that the missing piece is amplitude and bias rather than ranking. Cold-air pooling and topographic position remain unexamined covariates of both the observation and its macro reference, which is the forcing file itself. We present this as a consistency check against the mechanistic attribution rather than an attribution in its own right. An observational correlation cannot override the attribution the perturbation design isolates. The collinearity trap that motivated Chapter 1 applies to our follow-up analyses as much as to the literature. The ecological reading is also two-sided, since warmer nights under dense canopy do not relieve accumulated water stress. The nighttime regime is the main question this thesis leaves open.

D.7 Perspectives

D.7.1 Spaceborne LiDAR as the missing temporal dimension

The General Introduction set GEDI aside on the grounds that its sampling geometry and geolocation uncertainty keep it short of what plot-scale mechanistic modelling requires. That judgement stands, and our exploratory analysis confirms its core (Appendix G). GEDI's dense-canopy skill is real at its footprints but does not regress beyond them, so it cannot replace ALS as a per-plot structural source. GEDI can meet a weaker requirement, one this thesis left unmet. Every LiDAR quantity in the preceding chapters is a summer snapshot; no LiDAR truth existed outside that window. GEDI samples year-round (Dubayah et al., 2020) and supplies exactly that. Monthly composites at constant ALS-measured structure trace a full deciduous cycle, with a winter floor that is plausibly wood and branch area, so the sensor separates leaf from wood by season. GEDI canopy-cover profiles show the same seasonality elsewhere, concentrated in the 10 to 25 m layers of broadleaved stands and absent from coniferous ones (Liu et al., 2024). Their seasonal amplitude still ranks canopy density where the summer optical signal is flat. Their end of season falls 25 to 34 days later than the Sentinel-2 series, replicating at one site the continental gap reported by Cotrina-Sanchez et al. (2026). Chapter 3 confined its logger validation to summer, but the archive extends through autumn. We re-ran the Chapter 3 machinery over October and November, with the autumn limb of the seasonal series constrained by GEDI. That improved the simulated microclimate against those loggers (47 series plots; autumn ΔRMSE -0.08 °C, ΔR^2 $+0.19$). Composing the GEDI correction with the Chapter 3 combination gave the best forcing this thesis has evaluated, ahead of the combination alone in both summer and autumn (Appendix G, §G.5). Chapter 3 cannot itself decide that gain. On a full-leaf summer plateau the combination carries almost no seasonal shape and is close to the fixed LiDAR forcing by construction, so only the autumn limb tests a seasonal correction. These are exploratory, single-site results. The corrected series was reached through sequential variants scored on the same logger network, so the confidence intervals are conditional on that selection. We weight the large, pre-specified effects, September and autumn. The three sensors divide the roles. ALS supplies the magnitude and the dense-canopy ranking. Sentinel-2 supplies the open-canopy ranking and the per-plot leaf-on timing, which a site-level GEDI shape cannot take over, although Chapter 3 found that this timing adds little on the summer plateau. GEDI supplies the seasonal constraint at leaf-out and senescence, where optical senescence runs early (Oliveira & Zhang, 2025; Soudani et al., 2008). The late GEDI end of season mixes true leaves, marcescent leaves (typical of *Quercus* and *Fagus*), and exposed wood. For radiation and microclimate this structural signal is the relevant one, as the logger test supports. GEDI's dense-canopy skill does not regress beyond its footprints, so it complements rather than replaces the wall-to-wall products of Chapters 2 and 3 (Schwartz et al., 2023). Within about 150 m of a power-beam footprint, the direct GEDI measurement recovers 80 to 90 % of the ALS microclimate correlation ($n = 11$ – 18 plots). The growing footprint archive can therefore serve as local truth between airborne campaigns, at least in lowland broadleaf stands. At the montane site the footprint-level agreement with ALS is itself weak ($r = 0.13$ – 0.22 , September only, conifer admixture; Appendix G, §G.9), so no such check is available there. That boundary is part of the perspective.

D.7.2 An extreme-year stress test

Whether structural buffering persists under the climate it is supposed to mitigate is the operational premise of mapping forest microrefugia (Davis et al., 2019; De Frenne et al., 2019; Lenoir et al., 2017). The logger network stayed active through the record June and July 2022 heatwaves, and the answer at this site is encouraging on the axis that carries the evidence (Appendix G, §G.6).

The observed LAI-buffering coupling held on the hottest days: $r = -0.88$ in both years' hot windows, against -0.92 over all of summer 2021. The mean buffering slope did not collapse either (0.93 on the hottest 2022 days against 0.86 on the hottest 2021 days; Appendix G, Table G4). GEDI composites showed the 2022 canopy itself intact. The mean offset was deeper on hot days: -1.22 °C on the hottest 2022 days, against -1.13 °C on the hottest 2021 days and -0.73 °C over all of summer 2021. That deepening is the expected arithmetic of a sub-unity slope under hotter skies, not evidence of adaptation. The coupling itself did not move.

The scenario ranking also transferred out of its calibration year, with the fusion products keeping the lead. Part of the drought literature reports buffering that weakens as soils dry (Greiser et al., 2024). A lowland oak forest on deep soils in one exceptional year nuances that expectation rather than overturning it. The model's warm bias grew with heat severity ($+1.16$ °C), so the stress test supports the forcing and confirms the model bias of §D.6.

D.7.3 The vernal window and the direction of the association

Two small analyses close the loop between phenology and microclimate that recent work has opened from the other side (Wu et al., 2024). Layer-resolved GEDI profiles show the understory greening up about a week before the overstory, a measured vernal window. Crossing per-plot green-up dates with the plots' own spring temperatures then quantifies its thermal consequence. Later-leafing plots accumulate a warmer understory spring ($r = +0.55$, partialled on summer leaf area $+0.40$; $n = 33$). Green-up was dated by the same smoothed half-amplitude threshold as Appendix G, Table G1. The reciprocal association is weaker: winter understory temperature relates to green-up mostly through leaf area ($r = +0.40$, partial $+0.19$; $n = 35$; Appendix G, §G.7). The design is cross-sectional, and April canopy openness, species composition, and topographic position are not controlled. We therefore state the result as an asymmetry of associations. At stand scale, in one year, the spring understory temperature follows phenology more closely than winter temperature anticipates it. The asymmetry is testable further with the same archive, and it matters for any attempt to force spring microclimate from remotely sensed leaf-out dates.

D.7.4 The remaining avenues

The remaining avenues are listed in decreasing order of readiness. The first is dimensional. §D.6 locates the next margin of progress in the column itself. The archetype-dependent forcing of Chapter 3 should next be evaluated in the coupled canopy-atmosphere formulations named in the General Introduction (Grulois, 2026,

PhD thesis, ISPA Bordeaux, coupled ARPS-MuSICA; Van De Walle et al. (2026)), in the edges and gaps a column cannot see. Sequential data assimilation of the Sentinel-2 stream remains the one untested fusion family, and it combines naturally with a GEDI-tested phenology. A second, spectral avenue is on the retrieval side. Chapter 3 ran the Sentinel-2 inversion on the 10 m grid to match the microclimate resolution, which resamples the red-edge and short-wave-infrared bands from their native 20 m. Those bands carry much of the leaf-area information least affected by saturation. A retrieval at the native 20 m resolution, super-resolved back to 10 m, could reduce the dense-canopy saturation that drives the choice of sensor. No such avenue is needed in open canopy. Chapter 3 found the open plots to be genuine gaps: the Sentinel-2 signal there is understory and soil background (not a sub-2 m canopy layer hidden by the LiDAR height filter). Dedicated field LAI campaigns, with the standardised protocols their own uncertainties demand (Woodgate et al., 2015), would convert consistency statements into accuracy statements. Conifer and mixed stands are the stated boundary of every chapter, and the montane site of this thesis, where the height gate and GEDI both weaken, is the natural stress case. One practical note for replication. Subsampling the logger network shows that 15 to 20 well-spread loggers reproduce the slope-LAI coupling within this network (Appendix G, §G.8). Whether that transfers to another site is untested, but the validation design of this thesis looks reproducible at modest cost.

D.8 Concluding remarks

This thesis produces the specification a wall-to-wall microclimate map would be built on. The General Introduction set that boundary in advance. Leaf quantity is the controlling structural variable, and it has to be measured without saturation where the buffering is deepest. That still means airborne LiDAR wherever the canopy is dense. Which leaf-area source to trust then depends on the canopy archetype. The seasonal shape has to be constrained at leaf-out and senescence, outside the summer window Chapter 3 tested. A simulation bias remains that no forcing repairs, and that is where the next methodological work belongs.

The specification is falsifiable and cheap to test elsewhere: a network of 15 to 20 loggers and one summer. Its first two exploratory trials, one out-of-year (the 2022 heatwaves) and one against an independent sensor (the GEDI-corrected phenology), both came back positive.

Bouwen, K. (2025). *Interactions entre la structure de la canopée et le microclimat du sous-étage : implications pour la régénération des forêts* [PhD thesis]. Université de Bordeaux.

Cotrina-Sanchez, A., Coomes, D. A., Ball, J., Holcomb, A., Valentini, R., & Vaglio Laurin, G. (2026). The phenology of European forests as seen by MODIS Leaf Area Index and GEDI Plant Area Index: Toward an integrated approach. *Forest Ecology and Management*, 611, 123689. <https://doi.org/10.1016/j.foreco.2026.123689>

Davis, K. T., Dobrowski, S. Z., Holden, Z. A., Higuera, P. E., & Abatzoglou, J. T. (2019). Microclimatic buffering in forests of the future: The role of local water balance. *Ecography*, 42(1), 1–11. <https://doi.org/10.1111/ecog.03836>

- De Frenne, P., Zellweger, F., Rodríguez-Sánchez, F., Scheffers, B. R., Hylander, K., Luoto, M., Vellend, M., Verheyen, K., & Lenoir, J. (2019). Global buffering of temperatures under forest canopies. *Nature Ecology & Evolution*, 3(5), 744–749. <https://doi.org/10.1038/s41559-019-0842-1>
- Dubayah, R., Blair, J. B., Goetz, S., Fatoyinbo, L., Hansen, M., Healey, S., Hofton, M., Hurtt, G., Kellner, J., Luthcke, S., Armston, J., Tang, H., Duncanson, L., Hancock, S., Jantz, P., Marselis, S., Patterson, P. L., Qi, W., & Silva, C. (2020). The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography. *Science of Remote Sensing*, 1, 100002. <https://doi.org/10.1016/j.srs.2020.100002>
- Fujiwara, T., Marod, D., Thinkamphaeng, S., Phumphuang, W., Nakmuenwai, P., Kamyo, T., Brede, B., Naito, H., & Hosoi, F. (2026). Evaluating the sensitivity of Sentinel-2 reflectance to vertical canopy profiles using a voxel-based point density proxy framework. *International Journal of Remote Sensing*, 1–24. <https://doi.org/10.1080/01431161.2026.2720316>
- Greiser, C., Hederová, L., Vico, G., Wild, J., Macek, M., & Kopecký, M. (2024). Higher soil moisture increases microclimate temperature buffering in temperate broadleaf forests. *Agricultural and Forest Meteorology*, 345, 109828. <https://doi.org/10.1016/j.agrformet.2023.109828>
- Lang, N., Jetz, W., Schindler, K., & Wegner, J. D. (2022). *A high-resolution canopy height model of the Earth* (arXiv:2204.08322). arXiv. <https://arxiv.org/abs/2204.08322>
- Lenoir, J., Hattab, T., & Pierre, G. (2017). Climatic microrefugia under anthropogenic climate change: Implications for species redistribution. *Ecography*, 40(2), 253–266. <https://doi.org/10.1111/ecog.02788>
- Liu, X., Forkel, M., & Kranz, J. (2024). Investigating the Association of Seasonal Dynamics in GEDI Canopy Cover Profiles and Sentinel-1 Backscatter in Temperate Forests. *IEEE Geoscience and Remote Sensing Letters*, 21, 1–5. <https://doi.org/10.1109/LGRS.2024.3443525>
- Oliveira, P. V. C., & Zhang, X. (2025). Upper canopy and understory phenology of Brazilian Amazon forests seen by GEDI lasers. *Environmental Research Letters*, 20(4), 044015. <https://doi.org/10.1088/1748-9326/adbfa4>
- Potapov, P., Li, X., Hernandez-Serna, A., Tyukavina, A., Hansen, M. C., Kommareddy, A., Pickens, A., Turubanova, S., Tang, H., Silva, C. E., Armston, J., Dubayah, R., Blair, J. B., & Hofton, M. (2021). Mapping global forest canopy height through integration of GEDI and Landsat data. *Remote Sensing of Environment*, 253, 112165. <https://doi.org/10.1016/j.rse.2020.112165>
- Reek, J. E., Crowther, T. W., Lauber, T., Schemm, S., Parastatidis, D., Chrysoulakis, N., Huang, M., Piao, S., Zohner, C. M., & Smith, G. R. (2025). Forest edges are globally warmer than interiors and exceed optimal temperatures for vegetation productivity. *Communications Earth & Environment*, 6(1), 635. <https://doi.org/10.1038/s43247-025-02626-1>
- Schwartz, M., Ciais, P., De Truchis, A., Chave, J., Ottlé, C., Vega, C., Wigneron, J.-P., Nicolas, M., Jouaber, S., Liu, S., Brandt, M., & Fayad, I. (2023). *FORMS: Forest Multiple Source height, wood volume, and biomass maps in France at 10 to 30 m resolution based on Sentinel-1, Sentinel-2, and GEDI data with a deep learning approach* [Preprint]. ESSD - Land/Land Cover and Land Use. <https://doi.org/10.5194/essd-2023-196>
- Soudani, K., Le Maire, G., Dufrêne, E., François, C., Delpierre, N., Ulrich, E., &

- Cecchini, S. (2008). Evaluation of the onset of green-up in temperate deciduous broadleaf forests derived from Moderate Resolution Imaging Spectroradiometer (MODIS) data. *Remote Sensing of Environment*, 112(5), 2643–2655. <https://doi.org/10.1016/j.rse.2007.12.004>
- Terando, A. J., Youngsteadt, E., Meineke, E. K., & Prado, S. G. (2017). Ad hoc instrumentation methods in ecological studies produce highly biased temperature measurements. *Ecology and Evolution*, 7(23), 9890–9904. <https://doi.org/10.1002/ece3.3499>
- Van De Walle, E., Meunier, F., De Hertog, S. J., Terryn, L., Sanczuk, P., Calders, K., Wyffels, F., De Frenne, P., Stock, M., & Verbeeck, H. (2026). *ForEdgeClim* v1.0: A 3D process-based microclimate model incorporating vertical and lateral radiative and thermal fluxes to simulate forest edge-to-core transitions. *Geoscientific Model Development*, 19(10), 4661–4702. <https://doi.org/10.5194/gmd-19-4661-2026>
- Woodgate, W., Jones, S. D., Suarez, L., Hill, M. J., Armston, J. D., Wilkes, P., Soto-Berelov, M., Haywood, A., & Mellor, A. (2015). Understanding the variability in ground-based methods for retrieving canopy openness, gap fraction, and leaf area index in diverse forest systems. *Agricultural and Forest Meteorology*, 205, 83–95. <https://doi.org/10.1016/j.agrformet.2015.02.012>
- Wu, X., Niu, C., Liu, X., Hu, T., Feng, Y., Zhao, Y., Liu, S., Liu, Z., Dai, G., Zhang, Y., Van Meerbeek, K., Wu, J., Liu, L., Guo, Q., & Su, Y. (2024). Canopy structure regulates autumn phenology by mediating the microclimate in temperate forests. *Nature Climate Change*. <https://doi.org/10.1038/s41558-024-02164-2>